

March 13, 2025

Suzanne H. Plimpton,
Reports Clearance Officer,
National Science Foundation
2415 Eisenhower Avenue,
Alexandria, VA 22314

Re: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan (NSF_FRDOC_0001-3479)

Dear Ms. Plimpton,

The Autonomous Vehicle Industry Association (“AVIA”) writes in response to the National Science Foundation Networking and Information Technology Research and Development National Coordination Office’s Request for Information (“RFI”) on the “Development of an Artificial Intelligence (AI) Action Plan,” which was submitted on behalf of the Office of Science and Technology Policy (“OSTP”).¹ As an organization focused on the development and deployment of autonomous vehicles (“AVs”), AVIA would encourage the administration to include considerations for AVs as part of its AI Action Plan. AVs represent a key AI-enabled technology and an industry in which the U.S. continues to lead the world. Maintaining and growing U.S. leadership in AVs will depend on the creation of a federal policy framework that will help accelerate the safe deployment of the technology across the country.

As an organization, AVIA is committed to bringing the tremendous safety and mobility benefits of AVs—otherwise known as SAE Levels 4- and 5-capable vehicles—to consumers in a safe, responsible, and expeditious manner. AVIA’s membership is comprised of the world’s leading technology, automotive, ridesharing, trucking, and transportation companies.² Vehicles operated by AVIA members have driven more than 70 million autonomous miles on U.S. public roads, a distance roughly equivalent to 293 trips to the moon or driving across Route 66 over 29,000 times.³ AVs will play a pivotal role in addressing critical challenges facing our nation, including reducing the persistent and unacceptable level of traffic fatalities in our country, expanding transportation access, enhancing supply chain efficiency, and expanding economic output.

¹ Request for Information on the Development of an Artificial Intelligence (AI) Action Plan, 90 Fed. Reg. 9088 (Feb. 6, 2025). This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

² Our members include Aurora, Bot Auto, Cavnu, Cruise, Discount Tire, FedEx, Ford, Gatik, International, J.D. Power, Kodiak, Lyft, Motional, Nuro, Stack, Tier IV, Torc Robotics, Uber, UPS, Volkswagen Group of America, Volvo Cars, Volvo Autonomous Solutions, Waabi, Waymo, and Zoox. *See Our Mission and Members*, AUTONOMOUS VEHICLE INDUS. ASS’N, <https://theavindustry.org/> (last visited Mar. 13, 2025).

³ *Autonomous Vehicle Industry Association Releases First-Ever “State of AV” Report*, AUTONOMOUS VEHICLE INDUS. ASS’N (Apr. 10, 2024), <https://theavindustry.org/newsroom/press-releases/first-ever-state-of-av-report>.

The continued development and deployment of AVs in the United States will help revive our industrial capacity, create new manufacturing jobs, reduce supply chain costs, and ensure that the United States retains its international leadership and competitive advantage as this critical technology continues to grow and evolve. To fully realize these benefits, a supportive and uniform nationwide federal policy framework is essential. With this in mind, AVIA recently released *Securing American Leadership in Autonomous Vehicles*,⁴ a comprehensive set of federal policy recommendations that would accelerate the safe and timely deployment of AV technology and solidify the U.S. as the global leader in this transformational field. In the last several years, U.S. states have raced ahead on AV policy, and today 25 states covering 56 percent of our population have AV deployment statutes. While AVIA appreciates this interest by states, it is essential for the U.S. Department of Transportation to set nationally applicable design, construction, and performance standards for AVs, which is an authority it—and it alone—possesses. A federal framework for AVs is also vital to help maintain U.S. leadership in AV technology, as legal frameworks for AVs are already in place in China, the European Union, Germany, the United Kingdom, and elsewhere, giving foreign AV developers a level of regulatory certainty that can give them a competitive advantage over U.S. companies.

AVIA envisions a federal framework for AV safety that focuses on several key areas, including:

- **AV Safety, Transparency, and Accountability**

Actions that can be taken to promote AV safety, transparency, and accountability include undertaking a new rulemaking by the National Highway Traffic Safety Administration (“NHTSA”) to create new Federal Motor Vehicle Safety Standards (“FMVSS”) that apply to autonomous driving systems (“ADS”). These FMVSS should include behavioral competency tests to allow ADS manufacturers to self-certify their systems to demonstrate a basic level of ADS proficiency appropriate for the ADS’s operational design domain (“ODD”), and a requirement that ADS manufacturers develop a detailed “safety case” for their ADS. NHTSA should further establish a National AV Safety Data Repository, which will include relevant safety data about AV incidents and be available to state regulators, with appropriate protections for confidential business information. Such actions would provide additional regulatory certainty for AV developers, allowing them to innovate further and help grow the domestic AV industry.

- **Advancing American Leadership on AVs**

U.S. leadership in the global AV marketplace can be fostered by modernizing existing FMVSS to clarify that requirements for manual controls and certain indicators and telltales do not apply to Level 4 or Level 5 ADS-dedicated vehicles. These actions would support AV innovation and ease the expansion of AV deployments nationwide.

⁴ AUTONOMOUS VEHICLE INDUS. ASS’N, *SECURING AMERICAN LEADERSHIP IN AUTONOMOUS VEHICLES* (2025), <https://theavindustry.org/resources/Securing%20American%20Leadership%20in%20Autonomous%20Vehicles.pdf>.

- **Supporting Supply Chain Resiliency Through AV Trucking**

Autonomous trucking can ease supply chain burdens while supporting trucking jobs and increasing roadway safety. The Federal Motor Carrier Safety Administration (“FMCSA”) can support autonomous trucking by codifying the agency’s 2018 interpretation that the Federal Motor Carrier Safety Regulations do not require a human driver to operate or be present in a commercial motor vehicle (“CMV”) being operated by a Level 4 or Level 5 ADS. The agency should also update existing hours of service and inspection requirements that currently require action by a human driver, and, in partnership with industry and law enforcement, build on and support the existing consensus approach to autonomous truck inspection protocols.

Additionally, FMCSA should revisit the autonomous trucking industry’s petition to support new emergency warning device solutions that utilize cab-mounted beacons instead of driver placed devices, which the agency denied at the end of 2024. The petition is especially important as it presents a new safety technology that could be applied not only to AVs, but traditional CMVs, improving roadway safety for all users.

AVIA lays out these actions and others in greater detail in *Securing American Leadership in Autonomous Vehicles*, including actions that Congress can take to support AV innovation through comprehensive AV legislation. AVIA encourages OSTP to take these ideas and the needs of the AV industry into account as it continues to develop the AI Action Plan. Federal action is needed to help unlock the full potential of AVs and ensure that the United States remains the world leader in AV technology.

AVIA is grateful for the opportunity to provide these comments and welcomes the opportunity to engage further with OSTP on this and other matters. If there is anything further we can do to assist you or your staff, please do not hesitate to reach out.

Sincerely,

Ariel S. Wolf
General Counsel
Autonomous Vehicle Industry Association